

■ TerraSpectra (Project 1): Professional Engineering & Research Master Blueprint

Enterprise Architecture, Spectroscopy Science, Data Acquisition & Full Module Guide

■ EXECUTIVE SUMMARY & SYSTEM OVERVIEW

TerraSpectra is an enterprise-grade, GPU-accelerated Precision Agriculture and Geospatial Computer Vision platform.

The Engineering Challenge

Conventional crop health monitoring systems utilize 3-channel RGB (Red, Green, Blue) or 4-channel Multispectral imagery (e.g., NDVI using Near-Infrared). These systems operate on a major flaw: **they can only detect crop diseases after leaves visually turn yellow or brown (chlorosis and necrosis)**. By the time visual symptoms appear to human eyes or standard 2D Convolutional Neural Networks (2D-CNNs), internal cellular structures have already collapsed, making harvest yield loss unrecoverable.

Furthermore, standard 2D-CNN architectures treat image input as flat 2D spatial matrices with a few color channels. They are fundamentally incapable of modeling the **3D spatial-spectral volume** of Hyperspectral Data Cubes, which contain hundreds of narrow, contiguous spectral bands.

The TerraSpectra Solution

TerraSpectra ingests **3D Hyperspectral Data Cubes** capturing **200+ contiguous spectral bands of light** across 400 nm – 2500 nm. By coupling **3D Convolutional Neural Networks (3D-CNNs)** with **Spectral-Attention Vision Transformers (ViT)**, TerraSpectra detects sub-clinical biochemical anomalies—such as subtle chlorophyll-a/b depletion, anthocyanin spikes, and foliar water stress—predicting fungal blight outbreaks **up to 3 weeks before any visible symptoms manifest on the leaves**.

■ SECTION 1: DOMAIN & PHYSICS RESEARCH (WHAT TO STUDY)

Every software engineer and data scientist working on this project must master the following core domain disciplines:

```
graph TD
    A["Hyperspectral Remote Sensing"] --> B["Electromagnetic Spectrum Physics"]
    A --> C["Plant Spectroscopy & Pathophysiology"]
```

```
A --> D["Radiometric & Atmospheric Correction"]
```

```
B --> B1["VIS (400-700 nm): Chlorophyll Pigments"]
```

```
B --> B2["NIR (700-1300 nm): Mesophyll Cell Structure"]
```

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B --> B3["SWIR (1300-2500 nm): Foliar Moisture & Lignin"]
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C --> C1["Red Edge Shift (680-750 nm): Early Blight Indicator"]
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C --> C2["Narrow-band Vegetation Indices: PRI, WBI, SIPI"]
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D --> D1["Top-of-Atmosphere (TOA) Radiance -> Surface Reflectance"]
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```
D --> D2["Water Vapor & CO2 Absorption Suppression"]
```

1.1 Electromagnetic Spectrum Physics for Plant Canopy Analysis

Hyperspectral Imaging (HSI) measures light reflectance across hundreds of narrow, contiguous spectral channels (typically 5 nm – 10 nm bandwidths).

- **Visible Spectrum (VIS: 400 nm – 700 nm):**
- **Blue (400 nm – 500 nm):** Absorbed strongly by Carotenoids and Chlorophyll-a/b.
- **Green (500 nm – 600 nm):** Reflection peak (~550 nm), giving healthy plants their green color.
- **Red (600 nm – 700 nm):** Chlorophyll-a absorption peak at 675 nm.
- **The Red Edge Region (680 nm – 750 nm):**
- The boundary between red absorption and Near-Infrared reflection.
- **Critical Disease Signature:** When fungal pathogens (e.g., *Phytophthora infestans*) infect a plant, cell wall damage causes a **leftward shift (blue shift)** of the Red Edge toward shorter wavelengths **weeks before foliar chlorosis is visible**.
- **Near-Infrared (NIR: 700 nm – 1100 nm):**
- Highly reflected by healthy leaf spongy mesophyll cells (45% – 50% reflectance).
- Fungal blight causes internal cell collapse, dramatically dropping NIR reflectance.
- **Short-Wave Infrared 1 (SWIR-1: 1100 nm – 1400 nm) & SWIR-2 (1400 nm – 2500 nm):**
- Absorbs light based on foliar moisture, protein, lignin, and cellulose content.
- Major atmospheric water vapor absorption troughs occur at 1450 nm and 1940 nm.

1.2 Radiometric Calibration & Atmospheric Noise Suppression

Satellites measure Top-of-Atmosphere (TOA) Radiance (L_{TOA}). Atmosphere contains water vapor (H_2O), carbon dioxide (CO_2), and aerosols that scatter and absorb photons.

- **Conversion to Surface Reflectance (ρ_{surface}):** Engineers must study radiative transfer models (such as 6S - *Second Simulation of a Satellite Signal in the Solar Spectrum* or FLAASH):

$$\rho_{\text{surface}} = \frac{\pi \cdot (L_{\text{sensor}} - L_{\text{path}})}{\tau_v \cdot (E_0 \cdot \cos(\theta_s) \cdot \tau_s + E_{\text{down}})}$$

SECTION 2: DATASETS, FILE FORMATS & ACQUISITION GUIDE

2.1 Standard Data File Formats

- 1. **HDF5 (.h5 / .hdf5):** Hierarchical Data Format. Stores 3D arrays as datasets alongside attributes (CRS, Bounding Box, Sensor Type).
- 2. **GeoTIFF (.tif / .tiff):** Raster format with GeoKeys specifying CRS and Affine Transform matrix mapping pixel \$(x, y)\$ to real GPS coordinates.
- 3. **ENVI Header Format (.hdr + .dat/.img):** ASCII header file specifying samples, lines, bands, interleave mode (BSQ, BIL, BIP), and wavelength values.

2.2 Open-Source Hyperspectral Satellite Data Sources

Sensor / Mission	Agency / Source	Bands	Resolution	Data Access Link
AVIRIS / AVIRIS-NG	NASA JPL	224 bands (380 – 2510 nm)	4 m – 20 m airborne	aviris.jpl.nasa.gov
EO-1 Hyperion	NASA / USGS	242 bands (357 – 2576 nm)	30 m spaceborne	earthexplorer.usgs.gov
PRISMA	ASI (Italian Space Agency)	240 bands (400 – 2500 nm)	30 m spaceborne	prisma.asi.it
EnMAP	DLR (German Aerospace)	224 bands (420 – 2450 nm)	30 m spaceborne	enmap.org
Indian HySI	ISRO	64 bands (400 – 950 nm)	500 m spaceborne	bhuvan.nrsc.gov.in

SECTION 3: MATHEMATICAL & ALGORITHMIC FORMULAS

3.1 Principal Component Analysis (PCA) Math

- 1. **Flatten 3D Tensor:** $C \in \mathbb{R}^{H \times W \times B} \rightarrow X \in \mathbb{R}^{N \times B}$ where $N = H \cdot W$.
- 2. **Mean Centering:** $\bar{X} = X - \mu$.
- 3. **Covariance Matrix:** $\Sigma = \frac{1}{N - 1} \bar{X}^T \bar{X} \in \mathbb{R}^{B \times B}$.
- 4. **Eigendecomposition:** $\Sigma V = V \Lambda \quad (\text{Eigenvalues } \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_B)$.
- 5. **Cumulative Explained Variance Ratio:** $\text{EVR}(k) = \frac{\sum_{i=1}^k \lambda_i}{\sum_{j=1}^B \lambda_j} \geq 0.985$.
- 6. **Projection:** $Z = \bar{X} V_k \in \mathbb{R}^{N \times k} \rightarrow C_{\text{reduced}} \in \mathbb{R}^{H \times W \times k}$.

3.2 Narrow-Band Vegetation Indices (VIs)

- **Photochemical Reflectance Index (PRI):** $\text{PRI} = \frac{R_{531} - R_{570}}{R_{531} + R_{570}}$
- **Water Band Index (WBI):** $\text{WBI} = \frac{R_{970}}{R_{900}}$
- **Structure Insensitive Pigment Index (SIPI):** $\text{SIPI} = \frac{R_{800} - R_{445}}{R_{800} - R_{680}}$

SECTION 4: FULL MODULES & LIBRARIES TO DOWNLOAD (WITH EXPLICIT "WHY")

Python Backend & Data Science Libraries

```
pip install rasterio h5py netCDF4 scikit-learn torch torchvision timm captum fastapi uvicorn pydantic scipy
```

Package Name	Installation Command	Technical Justification ("WHY It Is Needed")
rasterio	pip install rasterio	Geospatial Raster I/O: Reads GeoTIFF files, extracts Affine Transform matrices, handles Coordinate Reference Systems (CRS: EPSG:4326/3857), and performs bounding box spatial queries.
h5py	pip install h5py	HDF5 File Interface: Ingests 3D Hyperspectral Data Cubes (.h5 files from NASA AVIRIS/Hyperion), extracts spectral band metadata attributes, and performs slice indexing without loading full multi-gigabyte datasets into RAM.
netCDF4	pip install netCDF4	Network Common Data Form: Ingests Copernicus Sentinel satellite NetCDF files (.nc) containing multi-dimensional environmental and spectral array layers.
scikit-learn	pip install scikit-learn	Dimensionality Reduction & Metrics: Implements sklearn.decomposition.PCA and IncrementalPCA to compress 200+ spectral bands down to 32 components while retaining >98.5% variance. Also provides classification evaluation metrics (f1_score, roc_auc_score).
torch (PyTorch)	pip install torch	Deep Learning Framework: Implements GPU-accelerated 3D Convolutional layers (nn.Conv3d), custom DataLoaders, CUDA tensor math, and autograd backpropagation loops for early blight classification.

torchvision	<code>pip install torchvision</code>	Vision Transforms & Datasets: Handles tensor normalization, data augmentation, spatial transforms, and image tensor preprocessing pipelines.
timm	<code>pip install timm</code>	PyTorch Image Models: Provides pretrained Vision Transformer (ViT) and Swin Transformer backbones adapted for spectral self-attention feature extraction.
captum	<code>pip install captum</code>	Explainable AI (XAI): Implements Integrated Gradients and DeepLIFT to extract feature attribution scores across spectral wavelengths, proving <i>which exact nanometer bands</i> triggered the disease alert.
fastapi	<code>pip install fastapi</code>	High-Performance Async REST API: Web framework for the backend prediction server. Provides automatic OpenAPI/Swagger documentation, async endpoint handling, and Pydantic request validation.
uvicorn	<code>pip install uvicorn</code>	ASGI Web Server: Lightning-fast production server running the FastAPI application with multi-worker process management.
pydantic	<code>pip install pydantic</code>	Data Validation: Enforces strict type checking for API JSON payloads, coordinate bounding boxes, and metadata DTOs.
geopandas	<code>pip install geopandas</code>	Geospatial Vector Engine: Handles GeoJSON, Shapefiles, spatial joins, and GIS polygon intersection checks for farm field boundaries.
scipy	<code>pip install scipy</code>	Scientific & Spectral Algorithms: Used for Savitzky-Golay spectral smoothing filters, curve fitting, and optimization algorithms.
shap	<code>pip install shap</code>	Shapley Additive Explanations: Computes game-theoretic feature attributions for model predictions.

■ Node.js & React Frontend GIS Libraries

```
npm install react react-dom @deck.gl/core @deck.gl/layers @deck.gl/geo-layers @deck.gl/react mapbox-gl react
```

Package Name	Installation Command	Technical Justification ("WHY It Is Needed")
<code>react & react-dom</code>	<code>npm install react react-dom</code>	UI Library: Core React 18 component framework for building the user interface, reactive state management, and component lifecycle control.
<code>@deck.gl/core</code>	<code>npm install @deck.gl/core</code>	WebGL GIS Engine: High-performance WebGL-powered data visualization engine capable of rendering millions of geospatial pixels directly on GPU.
<code>@deck.gl/layers</code>	<code>npm install @deck.gl/layers</code>	Core GIS Layers: Provides <code>BitmapLayer</code> and <code>GeoJsonLayer</code> to overlay disease probability heatmaps directly on top of satellite basemaps.
<code>@deck.gl/geo-layers</code>	<code>npm install @deck.gl/geo-layers</code>	Advanced GIS Tile Layers: Provides <code>TileLayer</code> for rendering pyramid raster map tiles dynamically as the user zooms and pans across farm zones.
<code>@deck.gl/react</code>	<code>npm install @deck.gl/react</code>	Deck.gl React Wrapper: Bridges WebGL Deck.gl canvas rendering with React component state.
<code>mapbox-gl</code>	<code>npm install mapbox-gl</code>	Mapbox Web Engine: Renders 3D satellite topography, terrain elevation maps, and high-resolution global satellite imagery basemaps.
<code>react-map-gl</code>	<code>npm install react-map-gl</code>	React Wrapper for Mapbox: Syncs Mapbox viewport state (<code>longitude</code> , <code>latitude</code> , <code>zoom</code> , <code>pitch</code> , <code>bearing</code>) seamlessly with React components.
<code>plotly.js-dist-min & react-plotly.js</code>	<code>npm install plotly.js-dist-min react-plotly.js</code>	Spectral Curve Charting: Renders interactive 224-band continuous reflectance line charts in the UI sidebar when an analyst clicks on any farm pixel.
<code>lucide-react</code>	<code>npm install lucide-react</code>	UI Icons: Modern SVG icon library for GIS controls, alert badges, layer toggles, and status indicators.
<code>tailwindcss</code>	<code>npm install -D tailwindcss postcss autoprefixer</code>	Utility-First CSS: Rapid styling for dark-mode glassmorphism cards, responsive sidebars, and control panels.
<code>vite</code>	<code>npm install -D vite @vitejs/plugin-react</code>	Build Tool & Dev Server: Next-generation frontend build tool providing instant Hot Module Replacement (HMR) and optimized TypeScript bundling.

■ SECTION 5: 6-PHASE ENGINEERING EXECUTION ROADMAP

■ Phase 1: Research & Data Acquisition

1. Acquire AVIRIS/Hyperion sample datasets or generate synthetic 224-band HDF5 files.
2. Verify spectral reflectance curves (400 nm – 2500 nm) for healthy crops vs infected blight zones.
3. Obtain Mapbox API Access Token from mapbox.com.

■ Phase 2: Data Ingestion & Dimensionality Reduction

1. Implement `HyperspectralDataLoader` to parse `.h5` and `.tif` files with `h5py` and `rasterio`.
2. Implement `SpectralPCAReducer` using `sklearn.decomposition.PCA` to reduce 224 bands $\rightarrow$ 32 components (>98.5% variance retained).
3. Compute narrow-band Vegetation Indices (PRI, WBI, SIPI).

■ Phase 3: AI Model Engineering (PyTorch 3D-CNN + ViT)

1. Build PyTorch 3D-CNN spatial-spectral feature extractor accepting 5D tensors (Batch, 1, Components, Height, Width).
2. Integrate Vision Transformer (ViT) self-attention module to learn inter-band dependencies.
3. Train model on healthy vs infected spatial-spectral samples.

■ Phase 4: Explainability (XAI) & Validation Audit

1. Implement `Captum` Integrated Gradients to extract wavelength importance scores.
2. Confirm the model's top attribution weights align with the Red Edge shift (680 nm – 750 nm).

■ Phase 5: Production FastAPI Tiling API

1. Build sliding-window tiling algorithm ($256 \times 256 \times 32$ tensor chunks with 16px overlapping borders).
2. Wrap model in FastAPI async endpoints (`/api/v1/predict/tile`).

■ Phase 6: Interactive React 18 + Deck.gl 3D GIS Dashboard

1. Scaffold React 18 + Vite + TypeScript application.
2. Build `MapContainer.tsx` integrating Deck.gl `BitmapLayer` / `GeoJsonLayer` over Mapbox GL satellite basemap.
3. Add Plotly.js spectral reflectance chart sidebar and historical timeline slider.